

Foundations of Arithmetic Power Geometry

VI:

A Conditional APG–Arakelov Height Coupling Program

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Abstract

This paper develops a conditional APG–Arakelov height coupling program connecting Arithmetic Power Geometry (APG) deformation functionals with Arakelov intersection invariants associated with Frey curves. Earlier APG volumes introduced local entropy-controlled closure deformation, stabilized information-geometric formulations, integrated closure-defect functionals, coordinate-dependent damping, and discrete prime-space regularization. The remaining obstruction is the passage from APG deformation energy on a compactified modular curve $X_0(N)^*$ to a height lower bound for the associated Frey elliptic curve $E_{a,b,p}$. This paper corrects the weight notation by distinguishing the original Euclidean APG weights from Fermat-normalized weights, formulates the APG potential on a compact Riemann surface, proves smooth solvability of the APG Poisson problem, proves positivity and quadratic Dirichlet-energy scaling, and derives a conditional APG-to-Faltings lower bound under an explicit APG projection hypothesis. The main novel addition is the APG modular-efficiency invariant

$$\mathfrak{E}_{\text{APG}}(E) = \frac{\left[\mathcal{D}_p^*(a, b) H(\widetilde{W}) \right]^2}{\deg(\pi)},$$

which measures APG deformation energy per unit modular complexity. The paper proves that, assuming the projection hypothesis, the Faltings height is bounded below by this modular-efficiency invariant up to a logarithmic conductor error. The paper does not claim an independent proof of Fermat’s Last Theorem. Instead, it gives a precise conditional reduction and identifies the exact open projection and modular-efficiency problems needed for a future unconditional APG height-obstruction program.

Keywords: Arithmetic Power Geometry; Arakelov geometry; Fermat’s Last Theorem; Frey curve; Faltings height; modular degree; Dirichlet energy; Shannon entropy; Diophantine geometry; modular efficiency.

1 Introduction

Fermat’s Last Theorem was proved through the modularity of semistable elliptic curves and Ribet’s level-lowering theorem (Ribet, 1990; Taylor & Wiles, 1995; Wiles, 1995). The classical proof associates a hypothetical primitive solution of

$$a^p + b^p = c^p, \quad p > 2,$$

to a Frey curve and then shows that the resulting mod- p Galois representation cannot arise from a modular form at the required lowered level (Frey, 1986; Mazur, 1977; Serre, 1987). The Wiles–Taylor–Wiles proof is arithmetic, representation-theoretic, and modular in nature (Diamond, 1996; Ribet, 1994; Taylor & Wiles, 1995; Wiles, 1995).

Arithmetic Power Geometry (APG) approaches the same arithmetic landscape from a different direction. Instead of treating the exponent only as a fixed integer, APG studies exponent variation as a deformation process. Earlier APG work introduced a local closure defect at the Euclidean target (Akhtar, 2026a), an entropy-governed first-order expansion (Akhtar, 2026a), a stabilized information-geometric metric and coordinate-stability theory (Akhtar, 2026b), an integrated closure-defect functional and conditional height-obstruction program (Akhtar, 2026c), coordinate-dependent damping and scale-corrected invariants (Akhtar, 2026d), and a discrete prime-space regularization together with scale-weighted defect functionals (Akhtar, 2026e). These constructions are analytically meaningful, but by themselves they do not prove a global Diophantine obstruction.

The purpose of APG VI is to isolate the precise APG–Arakelov bridge that would be required to convert APG deformation energy into a genuine arithmetic height obstruction. The intended bridge is

$$\text{APG deformation energy} \longrightarrow \text{Arakelov self-intersection} \longrightarrow h_F(E_{a,b,p}).$$

This bridge is nontrivial because APG energy naturally lives on an analytic model over a compactified modular curve $X_0(N)^*$, while the Faltings height belongs to the elliptic curve $E_{a,b,p}$. The modular parametrization

$$\pi : X_0(N)^* \longrightarrow E_{a,b,p}$$

is therefore indispensable (Diamond & Shurman, 2005; Shimura, 1971; Wiles, 1995).

This paper makes three corrections and one new contribution. First, it corrects the

weight inconsistency by distinguishing the original Euclidean APG weights from the Fermat-normalized weights. Second, it formulates the APG Poisson equation on the compact Riemann surface $X_0(N)^*(\mathbb{C})$, avoiding noncompactness ambiguity at the cusps. Third, it treats low-entropy configurations as a classical Ribet–Wiles reduction rather than as a new APG theorem. Fourth, it introduces the APG modular-efficiency invariant and proves a conditional Faltings-height lower bound in terms of this invariant.

The paper is intentionally conditional. It does not prove Fermat’s Last Theorem, the *abc* conjecture, the Szpiro conjecture, modularity, or level lowering. Instead, it proves internal analytic APG theorems and reduces the possibility of an independent APG height obstruction to a precise projection problem and a modular-efficiency growth problem.

2 Scope and Conditional Status

This manuscript is a conditional research-program paper. It does not prove Fermat’s Last Theorem, the *abc* conjecture, the Szpiro conjecture, modularity, or Ribet’s level-lowering theorem. The unconditional proof of Fermat’s Last Theorem remains the classical modular proof of Ribet (1990), Taylor & Wiles (1995), and Wiles (1995).

The purpose of this paper is different: it constructs an APG energy framework, proves its internal analytic scaling laws, and isolates the exact additional arithmetic hypotheses required to turn APG deformation energy into a height obstruction. Thus every contradiction obtained in the final sections is conditional on the APG–Arakelov projection hypothesis and the APG modular-efficiency growth conjecture.

3 Classical Background

Arakelov geometry supplies an intersection-theoretic framework for arithmetic surfaces (Arakelov, 1974; Faltings, 1984; Soulé, 1992). Faltings height connects arithmetic intersection theory with the complexity of abelian varieties and elliptic curves (Faltings, 1983, 1984; Bombieri & Gubler, 2006; Hindry & Silverman, 2000). Modular curves, modular parametrizations, and Galois representations provide the classical architecture behind the modern proof of Fermat’s Last Theorem (Diamond & Shurman, 2005; Katz & Mazur, 1985; Mazur, 1977; Ribet, 1990; Serre, 1987; Shimura, 1971; Wiles, 1995). General algebraic-geometric and analytic foundations underlying the notation, regularity assumptions, and arithmetic-surface framework include Hartshorne (1977), Lang (2002), Atiyah and Macdonald (1969), Ahlfors (1979), Rudin (1976), and Whittaker and Watson (1927). Additional context on Arakelov heights, Diophantine approximation, elliptic curves, and modular-height phenomena is provided by Abbes (2000), Bugeaud (2004), Knapp (1992), and Gross and Zagier (1986). Let $X_0(N)$ denote the modular curve of

level N , and let $X_0(N)^*$ denote its compactification. In this paper, all analytic Poisson equations are placed on the compact Riemann surface $X = X_0(N)^*(\mathbb{C})$. This convention avoids spectral and boundary ambiguities that arise on the noncompact quotient before adding cusps. On the arithmetic side, one may work with a regular arithmetic model $\mathcal{X}$ over $\text{Spec } \mathbb{Z}$, with controlled bad fibers and cuspidal boundary contributions (Bosch et al., 1990; Deligne, 1987; Moret-Bailly, 1985; Zhang, 1993).

For a hypothetical primitive Fermat triple $a^p + b^p = c^p$, the associated Frey curve is

$$E_{a,b,p} : y^2 = x(x - a^p)(x + b^p).$$

Its discriminant is of the expected scale

$$|\Delta_E| \asymp (abc)^{2p},$$

up to powers of 2 and minimal-model corrections (Silverman, 1992; Bosch et al., 1990; Frey, 1986). Height-discriminant estimates then give upper bounds for $h_F(E)$ in terms of $\log |\Delta_E|$ (Bombieri & Gubler, 2006; Hindry & Silverman, 2000; Silverman, 1986; Vojta, 1987).

4 APG Weight Normalization

A key correction in APG VI is the separation of two different weight systems.

Definition 1 (Euclidean APG weights). *For fixed positive coordinates $a, b > 0$, define the Euclidean APG weights by*

$$W^{(2)} = (w_a^{(2)}, w_b^{(2)}), \quad w_a^{(2)} = \frac{a^2}{a^2 + b^2}, \quad w_b^{(2)} = \frac{b^2}{a^2 + b^2}.$$

The corresponding entropy is

$$H(W^{(2)}) = -w_a^{(2)} \log w_a^{(2)} - w_b^{(2)} \log w_b^{(2)}.$$

These are the weights used in the local APG deformation theory near the Euclidean exponent $n = 2$. They govern the first-order closure-defect expansion and the integrated local deformation profile.

The Euclidean weight system $W^{(2)}$ and the associated entropy functional $H(W^{(2)})$ were originally introduced in APG I and subsequently used throughout APG II–V as the baseline deformation coordinates (Akhtar, 2026a,b,c,d,e).

Definition 2 (Fermat-normalized weights). *Assume hypothetically that*

$$a^p + b^p = c^p$$

for an odd prime $p > 2$. Define the Fermat-normalized weights by

$$\widetilde{W} = (\widetilde{w}_a, \widetilde{w}_b), \quad \widetilde{w}_a = \frac{a^p}{c^p}, \quad \widetilde{w}_b = \frac{b^p}{c^p}.$$

Then $\widetilde{w}_a + \widetilde{w}_b = 1$, and

$$H(\widetilde{W}) = -\widetilde{w}_a \log \widetilde{w}_a - \widetilde{w}_b \log \widetilde{w}_b.$$

Remark 1 (No replacement of APG I–V weights). *The Fermat-normalized weights $\widetilde{W}$ do not replace the Euclidean weights $W^{(2)}$. The Euclidean weights govern APG deformation from the $n = 2$ baseline. The Fermat-normalized weights are introduced only after imposing a hypothetical Fermat relation, so that the arithmetic mass distribution of the Fermat equation can be measured on the unit simplex.*

5 The APG Defect and Scale-Weighted Source

For a prime exponent $p > 2$, define the discrete projective APG defect by

$$\mathcal{D}_p(a, b) = 1 - \left((w_a^{(2)})^{p/2} + (w_b^{(2)})^{p/2} \right).$$

For positive finite coordinates a, b , one has

$$0 < \mathcal{D}_p(a, b) < 1.$$

The exponent $p/2$ appears because the APG defect compares the Euclidean L^2 -normalized baseline with its induced L^p -deformation. Since the Euclidean target is centered at $n = 2$, the deformation from 2 to p naturally sends the normalized weights to the powers $(w_a^{(2)})^{p/2}$ and $(w_b^{(2)})^{p/2}$. Thus $\mathcal{D}_p(a, b)$ measures the loss of Euclidean projective closure under the prime-exponent deformation. The strict positivity follows from monotonicity of ℓ^q -norms and the fact that $w_a^{(2)}, w_b^{(2)} \in (0, 1)$ with $w_a^{(2)} + w_b^{(2)} = 1$.

The scale-weighted APG defect is defined by

$$\mathcal{D}_p^*(a, b) = \mathcal{D}_p(a, b) \log(abc).$$

This functional combines the dimensionless APG deformation defect with the arithmetic coordinate scale. The APG energy source used in this paper is

$$\alpha_{\text{APG}} = \mathcal{D}_p^*(a, b) H(\widetilde{W}).$$

Lemma 1 (Positivity of the APG source). *Assume $a, b, c \geq 1$, $p > 2$, $a^p + b^p = c^p$, and $0 < \widetilde{w}_a, \widetilde{w}_b < 1$. If $abc > 1$, then*

$$\alpha_{\text{APG}} > 0.$$

Proof. For positive finite a, b , the projective APG defect satisfies $\mathcal{D}_p(a, b) > 0$. Since $abc > 1$, $\log(abc) > 0$, so $\mathcal{D}_p^*(a, b) > 0$. Since $0 < \tilde{w}_a, \tilde{w}_b < 1$ and $\tilde{w}_a + \tilde{w}_b = 1$, binary Shannon entropy satisfies $H(\tilde{W}) > 0$ (Cover & Thomas, 2006; Shannon, 1948). Hence $\alpha_{\text{APG}} > 0$. $\square$

6 Classical Low-Entropy Reduction

Remark 2 (Classical Ribet–Wiles reduction). *Let $0 < \eta < \log 2$. If $H(\tilde{W}) < \eta$, the Fermat-normalized coordinate distribution is close to the boundary of the probability simplex. This regime is not treated as a new APG theorem. The appropriate obstruction is the already established Ribet–Wiles modular obstruction: modularity of the semistable Frey curve, combined with level lowering, eliminates the hypothetical Fermat configuration (Ribet, 1990; Taylor & Wiles, 1995; Wiles, 1995). Thus APG VI does not claim independent novelty in the low-entropy regime.*

7 Logical Dependency Map

The logical structure of the paper is as follows:

APG source positivity $\implies$ Poisson solvability $\implies$ quadratic energy scaling $\implies$ Arakelov metric variational

The conditional arithmetic part then proceeds through

APG–Arakelov projection hypothesis $\implies$ modular-efficiency lower bound $\implies$ conditional height contr

The only unproved arithmetic inputs are:

APG–Arakelov projection hypothesis and APG modular-efficiency growth conjecture.

All other analytic statements in the paper are proved internally.

8 Compactified Modular-Curve Setup

Let

$$X = X_0(N)^*(\mathbb{C})$$

be the compact Riemann surface associated with the compactified modular curve. Let $d\mu$ be a smooth probability-compatible volume form on X , and let

$$V_X = \text{vol}(X) = \int_X d\mu.$$

Let

$$\bar{\omega}_X = (\omega_{\mathcal{X}/\mathbb{Z}}, \|\cdot\|_{\text{can}})$$

denote the relative canonical sheaf on a chosen regular arithmetic model $\mathcal{X}$, equipped with a smooth canonical Arakelov metric. The use of $X_0(N)^*$ is essential: the Poisson equation below is solved on a compact surface, so the only obstruction to solvability is the zero-average condition on the source.

Let $\rho \in C^\infty(X)$ be a nonnegative density satisfying

$$\int_X \rho d\mu = 1.$$

Define the zero-average source

$$f_\rho = \rho - \frac{1}{V_X}.$$

Then

$$\int_X f_\rho d\mu = 0.$$

9 The APG Poisson Potential

Definition 3 (APG Poisson potential). *The APG potential u_{APG} is the unique mean-zero smooth solution of*

$$\Delta u_{\text{APG}} = \alpha_{\text{APG}} \left(\rho - \frac{1}{V_X} \right), \quad \int_X u_{\text{APG}} d\mu = 0,$$

where Δ is the positive Laplace–Beltrami operator on the compact Riemann surface X .

Theorem 1 (Smooth solvability of the APG Poisson problem). *There exists a unique smooth mean-zero solution $u_{\text{APG}} \in C^\infty(X)$.*

Proof. On a compact Riemann surface, the kernel of the Laplace operator on functions consists exactly of the constants. The equation

$$\Delta u = g$$

has a smooth solution if and only if $\int_X g d\mu = 0$. Here

$$g = \alpha_{\text{APG}} \left(\rho - \frac{1}{V_X} \right),$$

and the zero-average condition follows from the definition of f_ρ . Elliptic regularity gives smoothness. Imposing $\int_X u d\mu = 0$ removes the additive constant ambiguity and gives uniqueness (Iwaniec & Kowalski, 2004; Sarnak, 1990). $\square$

10 Quadratic APG Dirichlet-Energy Scaling

Define the APG Dirichlet energy by

$$\mathcal{E}_{\text{APG}} = \int_X |\nabla u_{\text{APG}}|^2 d\mu.$$

Theorem 2 (Quadratic APG energy scaling). *Let ψ be the unique mean-zero solution of*

$$\Delta\psi = \rho - \frac{1}{V_X}.$$

Then

$$u_{\text{APG}} = \alpha_{\text{APG}}\psi$$

and

$$\mathcal{E}_{\text{APG}} = C_X \left[\mathcal{D}_p^*(a, b) H(\widetilde{W}) \right]^2,$$

where

$$C_X = \int_X |\nabla\psi|^2 d\mu \geq 0.$$

If $\rho \neq 1/V_X$, then $C_X > 0$.

Proof. By linearity of the Poisson equation,

$$\Delta(\alpha_{\text{APG}}\psi) = \alpha_{\text{APG}} \left(\rho - \frac{1}{V_X} \right).$$

Both u_{APG} and $\alpha_{\text{APG}}\psi$ have mean zero, so uniqueness implies

$$u_{\text{APG}} = \alpha_{\text{APG}}\psi.$$

Therefore

$$\nabla u_{\text{APG}} = \alpha_{\text{APG}} \nabla\psi.$$

Hence

$$\mathcal{E}_{\text{APG}} = \int_X |\alpha_{\text{APG}} \nabla\psi|^2 d\mu = \alpha_{\text{APG}}^2 \int_X |\nabla\psi|^2 d\mu.$$

Substituting $\alpha_{\text{APG}} = \mathcal{D}_p^*(a, b) H(\widetilde{W})$ gives the stated identity. If $\rho \neq 1/V_X$, then ψ is nonconstant, so $\int_X |\nabla\psi|^2 d\mu > 0$. $\square$

11 APG Conformal Metric Variation

Define the conformally twisted APG metric by

$$\|\cdot\|_{\text{APG}} = \|\cdot\|_{\text{can}} e^{-u_{\text{APG}}}.$$

Let

$$\bar{\omega}_{\text{APG}} = (\omega_{\mathcal{X}/\mathbb{Z}}, \|\cdot\|_{\text{APG}}).$$

Proposition 1 (Normalized Deligne–Arakelov variation). *Fix the Arakelov normalization in which a conformal change $\|\cdot\| \mapsto \|\cdot\|e^{-u}$ on a smooth Hermitian line bundle over an arithmetic surface satisfies*

$$\widehat{c}_1(\bar{L}e^{-u})^2 = \widehat{c}_1(\bar{L})^2 + \frac{1}{4\pi} \int_X |\nabla u|^2 d\mu$$

whenever $\int_X u d\mu = 0$. Under this normalization,

$$\widehat{c}_1(\bar{\omega}_{\text{APG}})^2 = \widehat{c}_1(\bar{\omega}_X)^2 + \frac{1}{4\pi} \mathcal{E}_{\text{APG}}.$$

Proof. Deligne’s metric variation formula for arithmetic self-intersection contains a linear term in u and a quadratic Dirichlet term (Deligne, 1987; Faltings, 1984; Soulé, 1992). The imposed normalization $\int_X u_{\text{APG}} d\mu = 0$ eliminates the linear term. The remaining term is

$$\frac{1}{4\pi} \int_X |\nabla u_{\text{APG}}|^2 d\mu = \frac{1}{4\pi} \mathcal{E}_{\text{APG}}.$$

The conclusion follows. □

Remark 3. *Different Arakelov and Laplacian conventions may change the numerical coefficient. The structural content is the quadratic energy contribution, not the convention-dependent constant.*

12 The APG Projection Hypothesis

The central missing theorem is a projection inequality from the compactified modular curve to the Frey elliptic curve.

Hypothesis 1 (APG–Arakelov projection hypothesis). *Let*

$$\pi : X_0(N)^* \longrightarrow E_{a,b,p}$$

be the modular parametrization. There exists an absolute constant $C_0 > 0$, depending at most on the normalization of the arithmetic model and the metric convention, such that

$$\mathcal{E}_{\text{APG}} \leq 4\pi \deg(\pi) \widehat{c}_1(\bar{\omega}_E)^2 + C_0 \deg(\pi) \log N.$$

Equivalently, for a suitable constant $C_1 > 0$,

$$\deg(\pi) \widehat{c}_1(\bar{\omega}_E)^2 \geq \frac{1}{4\pi} \mathcal{E}_{\text{APG}} - C_1 \deg(\pi) \log N.$$

Remark 4 (Geometric meaning). *This hypothesis says that APG deformation energy on $X_0(N)^*$ can be pushed down to the Frey curve up to controlled ramification, cuspidal, and vertical-fiber errors. A possible proof would compare ω_X with $\pi^*\omega_E$ and a correction divisor*

$$\omega_X \sim \pi^*\omega_E + R_\pi + V_{\text{cusp}} + V_{\text{bad}},$$

where R_π is ramification, V_{cusp} is the cuspidal boundary contribution, and V_{bad} is the vertical bad-fiber contribution. The difficulty is to bound the resulting arithmetic intersection and metric-distortion errors by $O(\deg(\pi) \log N)$. This is not a standard theorem in the required APG form.

12.1 Heuristic Evidence for the APG–Arakelov Projection Hypothesis

The APG–Arakelov projection hypothesis asserts that APG deformation energy on $X_0(N)^*$ can be transferred to the Frey curve $E_{a,b,p}$, up to controlled logarithmic conductor error:

$$\mathcal{E}_{\text{APG}} \leq 4\pi \deg(\pi) \widehat{c}_1(\bar{\omega}_E)^2 + C_0 \deg(\pi) \log N.$$

Although this statement is not proved here, it is geometrically motivated by the canonical divisor comparison induced by the modular parametrization

$$\pi : X_0(N)^* \longrightarrow E_{a,b,p}.$$

Formally, one expects a decomposition of the form

$$\omega_X \sim \pi^*\omega_E + R_\pi + V_{\text{cusp}} + V_{\text{bad}},$$

where R_π denotes the ramification divisor, V_{cusp} denotes the cuspidal boundary contribution, and V_{bad} denotes the vertical bad-fiber contribution.

The ramification term is constrained by the classical Riemann–Hurwitz formula

$$K_X = \pi^*K_E + R_\pi.$$

Since K_E is trivial on a smooth elliptic curve over the complex fiber, the complexity of K_X is absorbed by the ramification divisor. This suggests that ramification contributes at most degree-weighted geometric error.

The cuspidal contribution arises because $X_0(N)$ is noncompact before compactification. Passing to $X_0(N)^*$ adds finitely many cusps. Classical modular-curve estimates indicate that cusp growth is controlled by the level structure, and logarithmic conductor terms are the natural scale for Arakelov boundary corrections.

The vertical bad-fiber contribution is also expected to be logarithmic in N , because

bad reduction is supported at primes dividing the conductor. Arakelov intersection theory and minimal-model theory typically convert such bad-fiber contributions into logarithmic discriminant or conductor terms (Deligne, 1987; Zhang, 1993; Bosch et al., 1990).

Consequently, one expects an error structure of the form

$$R_\pi + V_{\text{cusp}} + V_{\text{bad}} = O(\deg(\pi) \log N)$$

at the level of arithmetic intersection contributions.

This subsection is heuristic and does not constitute a proof of the APG–Arakelov projection theorem. Its purpose is to identify the geometric origin of the error terms and to explain why the conjectured projection bound has the correct arithmetic scale.

13 APG Modular Efficiency

The preceding projection hypothesis naturally suggests that modular degree should not be studied alone. The relevant quantity is APG energy per unit modular complexity.

Definition 4 (APG modular-efficiency invariant). *Define*

$$\mathfrak{E}_{\text{APG}}(E_{a,b,p}) = \frac{\left[\mathcal{D}_p^*(a,b) H(\widetilde{W}) \right]^2}{\deg(\pi)}.$$

This invariant is the main conceptual addition of APG VI. It combines four ingredients: APG projective defect, arithmetic scale, entropy of the hypothetical Fermat distribution, and modular parametrization complexity. It is not claimed to be a classical invariant already present in the literature; rather, it is proposed as a new APG invariant whose usefulness depends on whether it satisfies strong growth or rigidity properties.

Proposition 2 (Structural Bounds for APG Modular Efficiency). *Let*

$$\mathfrak{E}_{\text{APG}}(E) = \frac{\left[\mathcal{D}_p^*(a,b) H(\widetilde{W}) \right]^2}{\deg(\pi)}.$$

Then

$$0 < \mathfrak{E}_{\text{APG}}(E) \leq \frac{(\log 2)^2 [\mathcal{D}_p^*(a,b)]^2}{\deg(\pi)}.$$

Furthermore, if

$$H(\widetilde{W}) \geq \eta > 0,$$

then

$$\mathfrak{E}_{\text{APG}}(E) \geq \frac{\eta^2 [\mathcal{D}_p^*(a, b)]^2}{\deg(\pi)}.$$

Proof. By definition,

$$\mathfrak{E}_{\text{APG}}(E) = \frac{[\mathcal{D}_p^*(a, b)H(W_f)]^2}{\deg(\pi)}.$$

For a nondegenerate binary probability vector W_f , Shannon entropy satisfies

$$0 < H(W_f) \leq \log 2.$$

Since $\mathcal{D}_p^*(a, b) > 0$ and $\deg(\pi) \geq 1$, it follows that

$$\mathfrak{E}_{\text{APG}}(E) > 0.$$

Using $H(W_f) \leq \log 2$, we obtain

$$\mathfrak{E}_{\text{APG}}(E) \leq \frac{(\log 2)^2 [\mathcal{D}_p^*(a, b)]^2}{\deg(\pi)}.$$

If $H(W_f) \geq \eta > 0$, then $H(W_f)^2 \geq \eta^2$. Hence

$$\mathfrak{E}_{\text{APG}}(E) \geq \frac{\eta^2 [\mathcal{D}_p^*(a, b)]^2}{\deg(\pi)}.$$

This proves the stated structural bounds. □

Remark 5 (Interpretation of Proposition 2). *Proposition 2 shows that $\mathfrak{E}_{\text{APG}}$ is a positive, entropy-controlled, degree-normalized deformation invariant. In the balanced regime, its lower behavior is governed by APG scale energy divided by modular degree. Therefore the decisive problem is not whether the modular degree is small in isolation, but whether APG deformation energy grows faster than modular complexity.*

14 Conditional APG-to-Faltings Lower Bound

Theorem 3 (APG modular-efficiency lower bound). *Assume the APG–Arakelov projection hypothesis. Then there exist constants $A > 0$ and $B > 0$, depending only on the chosen analytic and Arakelov normalizations, such that*

$$h_F(E_{a,b,p}) \geq A \mathfrak{E}_{\text{APG}}(E_{a,b,p}) - B \log N - O(1).$$

More explicitly, under the normalization above,

$$h_F(E_{a,b,p}) \geq \frac{C_X}{48\pi} \frac{\left[\mathcal{D}_p^*(a,b)H(\widetilde{W})\right]^2}{\deg(\pi)} - C_2 \log N - O(1).$$

Proof. By the APG–Arakelov projection hypothesis,

$$\deg(\pi)\widehat{c}_1(\bar{\omega}_E)^2 \geq \frac{1}{4\pi}\mathcal{E}_{\text{APG}} - C_1 \deg(\pi) \log N.$$

Dividing by $\deg(\pi)$ gives

$$\widehat{c}_1(\bar{\omega}_E)^2 \geq \frac{1}{4\pi \deg(\pi)}\mathcal{E}_{\text{APG}} - C_1 \log N.$$

Hence there exist positive constants A and B , depending only on the chosen Arakelov normalization, such that

$$h_F(E) \geq \frac{A}{\deg(\pi)}\mathcal{E}_{\text{APG}} - B \log N - O(1).$$

Under the normalization

$$A_F = \frac{1}{12},$$

one recovers the more explicit form

$$h_F(E) \geq \frac{1}{48\pi \deg(\pi)}\mathcal{E}_{\text{APG}} - C_2 \log N - O(1).$$

By quadratic APG energy scaling,

$$\mathcal{E}_{\text{APG}} = C_X \left[\mathcal{D}_p^*(a,b)H(\widetilde{W})\right]^2.$$

Substitution yields

$$h_F(E_{a,b,p}) \geq \frac{C_X}{48\pi} \frac{\left[\mathcal{D}_p^*(a,b)H(\widetilde{W})\right]^2}{\deg(\pi)} - C_2 \log N - O(1),$$

which is the claimed modular-efficiency lower bound. $\square$

15 A Correct Coordinate-Capacity Bound

The APG domination argument cannot rely on the false claim that $\log(abc)$ must grow linearly in p . The following elementary bound is the correct safe statement.

Lemma 2 (Constant coordinate-capacity lower bound). *If $a, b \geq 2$ and $a^p + b^p = c^p$, then*

$$\log(abc) \geq 3 \log 2 + \frac{\log 2}{p}.$$

Proof. Since $a, b \geq 2$,

$$c^p = a^p + b^p \geq 2^p + 2^p = 2^{p+1}.$$

Taking p -th roots gives

$$c \geq 2 \cdot 2^{1/p}.$$

Therefore

$$abc \geq 2 \cdot 2 \cdot (2 \cdot 2^{1/p}) = 8 \cdot 2^{1/p}.$$

Taking logarithms yields

$$\log(abc) \geq 3 \log 2 + \frac{\log 2}{p}.$$

□

16 The Modular-Efficiency Growth Problem

The earlier modular-degree suppression hypothesis is too strong if stated directly as an upper bound on $\deg(\pi)$. Modular degrees usually grow with conductor or level, and classical estimates do not provide the required suppression in the APG form (Diamond & Shurman, 2005; Goldfeld, 1985; Iwaniec & Kowalski, 2004; Sarnak, 1990). APG VI therefore replaces degree-only suppression by a modular-efficiency growth problem.

Conjecture 1 (APG modular-efficiency growth conjecture). *In the balanced regime $H(\widetilde{W}) \geq \eta > 0$, every hypothetical primitive Fermat trajectory satisfies*

$$\mathfrak{E}_{\text{APG}}(E_{a,b,p}) = \frac{\left[\mathcal{D}_p^*(a, b) H(\widetilde{W}) \right]^2}{\deg(\pi)} \gg p \log(abc) + \log N.$$

Remark 6. *This conjecture is not proved in this paper. It is the central open problem isolated by the APG VI framework. It is weaker and more natural than studying modular degree alone because it compares modular complexity against the APG deformation energy that must pass through the modular parametrization.*

17 Conditional Height Contradiction

Theorem 4 (Conditional APG height obstruction). *Assume the APG–Arakelov projection hypothesis and the APG modular-efficiency growth conjecture. Then no balanced primitive*

Fermat triple can satisfy

$$a^p + b^p = c^p$$

for all sufficiently large odd primes p .

Proof. By the APG modular-efficiency lower bound,

$$h_F(E_{a,b,p}) \geq A \mathfrak{E}_{\text{APG}}(E_{a,b,p}) - B \log N - O(1).$$

By the modular-efficiency growth conjecture, the right-hand side eventually dominates $p \log(abc)$ in the balanced regime.

On the other hand, for the Frey curve,

$$\log |\Delta_E| \leq 2p \log(abc) + O(1),$$

up to minimal-model corrections at 2. Standard height-discriminant estimates imply

$$h_F(E_{a,b,p}) \leq \frac{1}{12} \log |\Delta_E| + O(1) \leq \frac{1}{6} p \log(abc) + O(1).$$

Thus the APG lower bound eventually exceeds the classical height-discriminant upper bound, a contradiction. Therefore no balanced primitive Fermat triple can exist under the two stated assumptions. $\square$

18 Main Conditional APG–Arakelov Result

Theorem 5 (Conditional APG–Arakelov obstruction theorem). *Assume:*

- (i) *the classical Ribet–Wiles obstruction for low-entropy configurations;*
- (ii) *the APG–Arakelov projection hypothesis;*
- (iii) *the APG modular-efficiency growth conjecture for balanced configurations.*

Then no nontrivial primitive Fermat triple exists for odd prime exponent $p > 2$.

Proof. Every Fermat-normalized distribution $\widetilde{W}$ lies either in a low-entropy regime $H(\widetilde{W}) < \eta$ or a balanced regime $H(\widetilde{W}) \geq \eta$. The low-entropy regime is eliminated by the classical Ribet–Wiles modular obstruction. The balanced regime is eliminated by the conditional APG height obstruction. Hence all cases are excluded under the stated assumptions. $\square$

19 Discussion

APG VI should be interpreted as a conditional height-coupling program, not as an unconditional proof of Fermat's Last Theorem. The unconditional proof of Fermat's Last Theorem is already known through the modular method (Ribet, 1990; Taylor & Wiles, 1995; Wiles, 1995). The value of APG VI is different: it identifies a possible entropy–deformation–height mechanism and isolates the exact theorem required to make that mechanism arithmetically decisive.

These results should be interpreted as the sixth stage of the APG program. Historically, APG I established the local entropy-governed deformation theorem, APG II introduced the stabilized information-geometric framework, APG III globalized the defect through the integrated closure-defect functional A_0 , APG IV introduced scale-corrected invariants and coordinate-dependent damping, APG V established the discrete prime-space formulation and scale-weighted defect functionals, and APG VI develops the conditional APG–Arakelov height-coupling program (Akhtar, 2026a,b,c,d,e). The internally proved results of APG VI are:

1. the correction and separation of Euclidean APG weights and Fermat-normalized weights;
2. positivity of the APG energy source;
3. smooth solvability of the APG Poisson potential on $X_0(N)^*(\mathbb{C})$;
4. quadratic Dirichlet-energy scaling;
5. conformal Arakelov metric variation under a fixed normalization;
6. the conditional APG modular-efficiency lower bound under the APG–Arakelov projection hypothesis.

The unproved frontier problems are:

1. the APG–Arakelov projection theorem, including ramification, cusp, vertical-fiber, and metric-distortion control;
2. the APG modular-efficiency growth conjecture.

The most important new object introduced here is

$$\mathfrak{E}_{\text{APG}}(E) = \frac{\left[\mathcal{D}_p^*(a, b) H(\widetilde{W}) \right]^2}{\deg(\pi)}.$$

This invariant reframes the earlier modular-degree suppression difficulty. Instead of asking for modular degree to be small in isolation, APG VI asks whether APG deformation

energy per unit modular complexity is forced to grow along hypothetical Fermat configurations. This formulation is more natural, more measurable, and more likely to survive mathematical scrutiny.

20 Conclusion

This paper develops a conditional APG–Arakelov framework linking Arithmetic Power Geometry deformation functionals with Arakelov intersection invariants associated with Frey elliptic curves. The formulation resolves the weight-normalization ambiguity by distinguishing the Euclidean APG weights from the Fermat-normalized weights, places the analytic construction on the compactified modular curve $X_0(N)^*$, and treats low-entropy configurations through the classical Ribet–Wiles modular obstruction rather than through a new APG theorem.

Within this framework, the paper proves positivity of the APG source term, smooth solvability of the associated Poisson problem, quadratic Dirichlet-energy scaling, and a normalized Arakelov metric-variation formula. It further introduces the APG modular-efficiency invariant $\mathfrak{E}_{\text{APG}}(E_{a,b,p}) = [\mathcal{D}_p^*(a,b)H(W_f)]^2 / \deg(\pi)$, which measures APG deformation energy relative to modular-parametrization complexity. Assuming the APG–Arakelov projection hypothesis, a conditional lower bound for the Faltings height is obtained in terms of this invariant.

The principal contribution of APG VI is the isolation of two precise arithmetic problems: the APG–Arakelov projection problem and the APG modular-efficiency growth problem. These identify the exact additional ingredients required for any future APG-based height-obstruction approach. Consequently, APG VI should be viewed as a conditional arithmetic-geometric research program that clarifies the relationship among deformation energy, entropy, modular complexity, and arithmetic height within the APG framework. It does not provide an unconditional proof of Fermat’s Last Theorem, whose proof remains the classical modular-theoretic work of Ribet, Taylor–Wiles, and Wiles.

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